Thus, $-\frac{1}{2} \leq \sin A \cos A \leq \frac{1}{2}$.

From this, we infer that the maximum value of $\sin A \cos A$ is $\frac{1}{2}$ when $A = \frac{\pi}{4}$.

Example 3.21 A foot ball player can kick a football from ground level with an initial velocity of 80 ft/second. Find the maximum horizontal distance the football travels and at what angle? (Take $g = 32$).

Solution:
The formula for horizontal distance R is given by

$$R = \frac{u^2 \sin 2\alpha}{g} = \frac{(80 \times 80) \sin 2\alpha}{32} = 10 \times 20 \sin 2\alpha.$$

Thus, the maximum distance is 200 ft.
Hence, he has to kick the football at an angle of $\alpha = 45^\circ$ to reach the maximum distance.

Identity 3.8: $\cos 2A = \cos^2 A - \sin^2 A$

Proof.

We know that $\cos(\alpha + \beta) = \cos\alpha \cos\beta - \sin\alpha \sin\beta$

Take $\alpha = \beta = A$. We have $\cos(A + A) = \cos A \cos A - \sin A \sin A$

$$\cos 2A = \cos^2 A - \sin^2 A.$$

From the identity $\cos 2A = \cos^2 A - \sin^2 A$, we also have

$$\cos 2A = \cos^2 A - \left(1 - \cos^2 A\right) = 2\cos^2 A - 1 \text{ and}$$
$$\cos 2A = (1 - \sin^2 A) - \sin^2 A = 1 - 2\sin^2 A.$$

Identity 3.9: $\tan 2A = \dfrac{2\tan A}{1 - \tan^2 A}$

Proof.

Now, $\tan(\alpha + \beta) = \dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha \tan\beta}$

Take $\alpha = \beta = A$. We have, $\tan(A + A) = \dfrac{\tan A + \tan A}{1 - \tan A \tan A}$

$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}.$$

Identity 3.10: $\sin 2A = \dfrac{2\tan A}{1 + \tan^2 A}$

Proof.

$$\text{we know that } \sin 2A = 2\sin A \cos A = \frac{2\sin A \cos A}{\sin^2 A + \cos^2 A}$$
$$= \frac{\frac{2\sin A \cos A}{\cos^2 A}}{\frac{\sin^2 A + \cos^2 A}{\cos^2 A}} = \frac{2\tan A}{1 + \tan^2 A}.$$